

Why Constrained Optimization?

In Section 14.7 we found absolute extrema by checking critical points **and** the boundary.

But what if the constraint is more complicated than a rectangular box?

Real-world examples:

- Maximize profit subject to a fixed budget
- Minimize material used to build a container with a required volume
- Find the hottest point on a wire bent into a curve

In each case we want to optimize $f(x, y)$ (or $f(x, y, z)$) but we are **restricted** to points satisfying a constraint $g(x, y) = k$.

Lagrange multipliers give us an elegant, systematic method for optimizing a function **subject to a constraint** — no parameterization needed.

A Motivating Example

Example 1

Find the absolute maximum and minimum of $f(x, y) = 6 - 2x + 0.1x^2 + 0.3xy$ on the ellipse $0.2x^2 + 0.3y^2 = 3.6$.

The **constraint** is that we only consider points (x, y) lying on the ellipse. We can't just find critical points of f — the extrema occur *on the ellipse*.



A 3D surface showing the function $f(x, y) = 6 - 2x + 0.1x^2 + 0.3xy$ restricted to an elliptical constraint curve. The curve on the surface reaches a highest and lowest point.

The surface $z = f(x, y)$ restricted to the ellipse | [Desmos 3D](https://www.desmos.com/3d) [https://www.desmos.com/3d]

Let's look at two approaches to this problem.

Approach 1: Parameterize the Constraint

Idea: Turn the constraint curve into a parametric curve, then substitute into f to get a single-variable problem.

The ellipse $0.2x^2 + 0.3y^2 = 3.6$ can be parameterized as:

$$x = \sqrt{18} \cos t, \quad y = \sqrt{12} \sin t, \quad 0 \leq t < 2\pi$$

Now substitute into f :

$$f(t) = f(x(t), y(t)) = 6 - 2\sqrt{18} \cos t + 0.1(18) \cos^2 t + 0.3\sqrt{18}\sqrt{12} \cos t \sin t$$

This is a **single-variable** calculus problem — find the absolute max/min on $[0, 2\pi)$.

Parameterization **works** but has drawbacks: finding a parameterization can be difficult, and the resulting function can be messy.

The Geometry Behind Lagrange Multipliers

Think about walking along the constraint curve $g(x, y) = k$. At most points, f is either increasing or decreasing as you walk.

At an **extremum** along the constraint, f is neither increasing nor decreasing — the level curves of f are **tangent** to the constraint curve.

Recall:

- ∇f is **perpendicular** to level curves of f
- ∇g is **perpendicular** to the constraint curve $g = k$

At a constrained extremum, ∇f and ∇g point in the **same (or opposite) direction** — they are **parallel**.

If ∇f is parallel to ∇g , then there exists a scalar λ such that:

$$\nabla f = \lambda \nabla g$$

This scalar λ is called the **Lagrange multiplier**.

Geometric intuition: When ∇f has a component along the constraint curve, you can still increase f by moving along the curve. Only when ∇f is entirely perpendicular to the curve (parallel to ∇g) have you reached an extremum.

The Method of Lagrange Multipliers

Definition 1: Method of Lagrange Multipliers

To find the maximum and minimum values of $f(x, y, z)$ subject to the constraint $g(x, y, z) = k$ (assuming these extreme values exist and $\nabla g \neq \vec{0}$ on the surface $g = k$):

1. Find all values of x, y, z , and λ such that

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \quad \text{and} \quad g(x, y, z) = k$$

1. Evaluate f at all the points (x, y, z) found in step 1. The largest value is the **maximum**; the smallest is the **minimum**.

In two variables, $\nabla f = \lambda \nabla g$ becomes the system:

$$f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad g(x, y) = k$$

That's **3 equations in 3 unknowns** (x, y, λ).

You often don't need the actual value of λ — it can be eliminated from the system to find x, y , and z .

Approach 2: Lagrange Multipliers on the Ellipse

Let's revisit Example 1 using Lagrange multipliers.

$$f(x, y) = 6 - 2x + 0.1x^2 + 0.3xy \quad \text{and} \quad g(x, y) = 0.2x^2 + 0.3y^2 = 3.6.$$

Step 1: Compute gradients.

$$\nabla f = \langle -2 + 0.2x + 0.3y, 0.3x \rangle$$

$$\nabla g = \langle 0.4x, 0.6y \rangle$$

Step 2: Set $\nabla f = \lambda \nabla g$ and $g = 3.6$:

$$-2 + 0.2x + 0.3y = 0.4\lambda x$$

$$0.3x = 0.6\lambda y$$

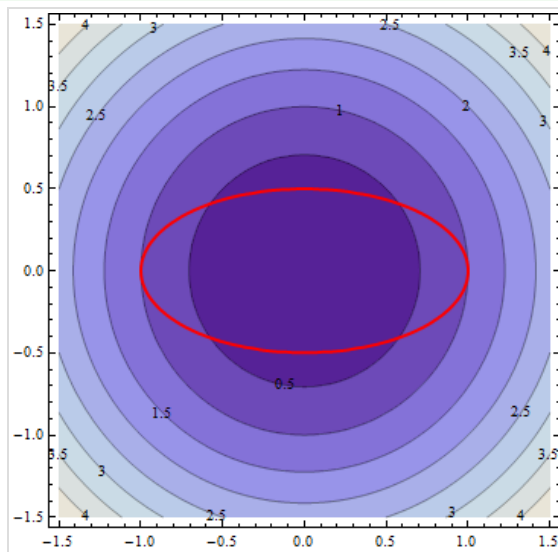
$$0.2x^2 + 0.3y^2 = 3.6$$

Solve this system. Hint: from the second equation, find λ in terms of x and y , then substitute.

Example 2: Closest and Farthest from the Origin

Example 2

Find the maximum and minimum values of $f(x, y) = x^2 + y^2$ on the ellipse $x^2 + 4y^2 = 1$.



The paraboloid restricted to the ellipse | [Desmos 3D](https://www.desmos.com/3d) [https://www.desmos.com/3d]

Set up: $f(x, y) = x^2 + y^2$, $g(x, y) = x^2 + 4y^2 = 1$.

Compute gradients: $\nabla f = \langle 2x, 2y \rangle$, $\nabla g = \langle 2x, 8y \rangle$.

Write the system from $\nabla f = \lambda \nabla g$ and $g = 1$. Solve for x , y , and λ .

Your Turn: Lagrange Practice

Example 3

Find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$.

Hint: Set up $\nabla f = \lambda \nabla g$ where $g(x, y) = x^2 + y^2$.

Set up the three equations, then solve for x , y , and λ .

Three Variables: $f(x, y, z)$ with Constraint $g(x, y, z) = k$

The method extends naturally to functions of three variables.

Optimize $f(x, y, z)$ subject to $g(x, y, z) = k$:

$$\nabla f = \lambda \nabla g \implies f_x = \lambda g_x, \quad f_y = \lambda g_y, \quad f_z = \lambda g_z$$

Plus the constraint: $g(x, y, z) = k$.

That's **4 equations in 4 unknowns** (x, y, z, λ) .

Same idea, one more variable. The system is bigger but the strategy is the same: use the gradient equations to relate the variables, then plug into the constraint.

Example 4: Minimum Surface Area Box

Example 4

Find the dimensions of a rectangular box with volume 1000 cm^3 that has the **smallest** surface area.

Set up the problem:

- Surface area: $f(x, y, z) = 2xy + 2xz + 2yz \leftarrow$ minimize this
- Volume constraint: $g(x, y, z) = xyz = 1000$

Compute gradients:

$$\nabla f = \langle 2y + 2z, 2x + 2z, 2x + 2y \rangle$$

$$\nabla g = \langle yz, xz, xy \rangle$$

Write the four equations and solve. Look for symmetry!

The cube minimizes surface area for a given volume. Symmetry in the equations often signals that the optimal solution is symmetric too.

When to Use Lagrange Multipliers

How does Lagrange compare with other optimization methods from Chapter 14?

Method	When to Use
Critical points (14.7)	No constraints; find local/absolute extrema on an open region
Closed & bounded (14.7)	Extrema on a region with boundary; check interior + boundary
Parameterize the constraint	Constraint has a simple parameterization (circle, line, etc.)
Lagrange multipliers (14.8)	Constraint is a curve or surface defined by $g = k$; no easy parameterization

Lagrange multipliers are most powerful when the constraint is hard to parameterize or when you have **multiple constraints**.

Homework

Section 14.8: 1-13 odd, 45